# Anatomy of a conversion factor

① **Starting quantity**: the number and the unit we want to change.

② **Numerator**: the new unit we want to obtain.

③ **The "1" always goes with the bigger unit**: 1 km = $10^3$ m.

$$2.5\ \cancel{\text{km}} \cdot \frac{10^3\ \text{m}}{1\ \cancel{\text{km}}} = 2.5 \cdot 10^3\ \text{m} = 2500\ \text{m}$$

④ **Denominator**: the old unit, so that it cancels out with the starting unit.

Top and bottom are the **same amount**, so the factor is worth exactly **1**: the quantity does not change, only its unit.